Jake Ryan

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Professional Summary

Robotics-focused engineer with a Computer Science background and hands-on experience across the full stack of a robotic system: perception, control, and embedded software. Comfortable working in ROS/C++ for real-time control as well as Python for tooling, simulation, and data analysis. Experienced running human-subject studies and iterating on hardware-software integration end to end.

Experience

Artificial Intelligence Research Assistant May 2019 – July 2019
Southwestern University *Georgetown, TX*

- Designed procedural generation algorithms for game environments, later applied to path-planning test scenarios
- Built a Java-based simulation environment to validate generated environments against agent navigation
- Ran a human-subject study and statistical analysis to evaluate environment quality (8-page paper, conference talk)
- Presented findings at the World Conference on Computational Intelligence

Undergraduate Research Assistant June 2020 – Present
Texas A&M University *College Station, TX*

- Built data pipelines (FastAPI, PostgreSQL) to log and analyze sensor/interaction data from research systems
- Containerized services with Docker to keep experiments reproducible across lab machines

Information Technology Support Specialist Sep. 2018 – Present
Southwestern University *Georgetown, TX*

- Maintained and repaired lab hardware, including embedded and classroom automation equipment
- Diagnosed hardware/software faults across 200+ devices and computer-controlled peripherals

Projects

Autonomous Line-Following Rover | *ROS, C++, Python, Raspberry Pi* 2021

- Built a ROS-based rover with a PID control loop for line-following using a camera and IR sensor array
- Implemented a basic obstacle-avoidance node using ultrasonic sensor fusion
- Tuned control parameters in simulation (Gazebo) before deploying to hardware

Simple Paintball Server Automation | *Java, Spigot API, Maven, TravisCI* May 2018 – May 2020

- Developed an event-driven plugin controlling real-time game state for 2K+ users
- Applied continuous integration to reliably ship updates to a live, always-on system

Technical Skills

Programming: C/C++, Python, Java, SQL
Robotics & Control: ROS/ROS2, Gazebo, PID control, path planning, sensor fusion (basic SLAM concepts)
Hardware: Raspberry Pi, Arduino, IR/ultrasonic sensors, basic embedded I/O
Tools: Git, Docker, Linux, VS Code, MATLAB (basic)
Data & Analysis: pandas, NumPy, Matplotlib, statistical study design

Education

Southwestern University Georgetown, TX
Bachelor of Arts in Computer Science, Minor in Business *Aug. 2018 – May 2021*

Blinn College Bryan, TX
Associate's in Liberal Arts *Aug. 2014 – May 2018*